Complexity analysis

$a(b+jc) = ab + jac$: 2 real multiplications

$(a+jb) + (c+jd) = (a+c) + j(b+d)$: 2 real additions

$(a+jb)(c+jd) = (ac-bd) + j(ad+bc)$: 4 real multiplications & 2 real additions

Trivial twiddles (No real multiplication or addition)

Let $w_8^r = e^{-j2\pi r/8}$

$w_8^0 = 1 \qquad w_8^4 = -1$

$w_8^2 = -j \Rightarrow (a+jb)(-j) = b-ja$: Swap real & imaginary parts & change sign

$w_8^6 = j \Rightarrow (a+jb)(j) = -b+ja$: Swap real & imaginary parts & change sign

8 - independent direct form channels

$$Y_k[m] = \sum_{n=0}^{31} h[n]\, e^{-j2\pi kn/8}\, x[8m+7-n]$$ Assuming $h[n]\, w_8^{kn}$ is precomputed

Counting twiddles, we have $w_8^0, w_8^k, w_8^{2k} \dots w_8^{31k}$ for a fixed k

$w_8^{kn} = w_8^{(k+n+8)} \Rightarrow$ we need to check only $w_8^0 \dots w_8^{7k}$

$1, -1, j, -j$ require no real multiplications $\Rightarrow w_8^0, w_8^2, w_8^4, w_8^6$ are trivial

$w_8^1, w_8^3, w_8^5, w_8^7$ are complex $\therefore h[n]x[n']w_8^{kn}$ will involve 2 real multiplication

For k = even, all 32 terms are trivial $\Rightarrow$ 32 real multiplications

For k = odd, 4 unique trivial twiddles $\Rightarrow$ 16 non trival total $\Rightarrow$ 48 real multiplications

$\therefore$ Total 320 real multiplications

For $k=0$, $w_8^0 = 1 \Rightarrow$ 31 real additions (similar for $k=4$)

For $k=1$, $w_8^0 = 1$, $w_8^2 = -j$, $w_8^4 = -1$, $w_8^6 = j \Rightarrow$ 46 real additions (similar for $k = 3, 5, 7$)

For $k=2$, 16 real & 16 imaginary terms $\Rightarrow$ 30 real additions (similar for $k=6$)

$\therefore$ total 306 real additions required per frame

Polyphase FIR w/ FFT

$$v_r[m] = \sum_{j=0}^{3} h[8j+r]\, x[8(m-j)+7-r] \quad \text{for } r = 0 \dots 7$$

Total 24 real additions & 32 real multiplications per frame

Now analyzing $Y_k(m) = \sum_{r=0}^{7} v_r[m]\, e^{-j2\pi kr/8}$ Using FFT

$$\Rightarrow Y_k[m] = \underbrace{\sum_{s=0}^{3} v_{2s}[m] W_4^{ks}}_{E_k} + W_8^k \underbrace{\sum_{s=0}^{3} v_{2s+1}[m] W_4^{ks}}_{O_k}$$

E_k: 4 point DFT of $v_0 v_2 v_4 v_6$ $\quad$ O_k: 4 point DFT of $v_1 v_3 v_5 v_7$

$Y_k = E_k + W_8^k O_k \quad Y_{k+4} = E_k - W_8^k O_k \quad k = 0, 1, 2, 3$

For 4 point DFT of $a_0\ a_1\ a_2\ a_3$

$s_0 = a_0 + a_2 \quad s_1 = a_1 + a_3 \quad d_0 = a_0 - a_2 \quad d_1 = a_1 - a_3$ Stage-1

$A_0 = s_0 + s_1 \quad A_1 = d_0 - jd_1 \quad A_2 = s_0 - s_1 \quad A_3 = d_0 + jd_1$ Stage-2

$k = 0 \quad Y_0 = E_0 + O_0 \quad Y_4 = E_0 - O_0$ 2 real additions

$k = 2 \quad Y_2 = E_2 - jO_2 \quad Y_6 = E_2 + jO_2$

$k = 1 \quad Y_1 = E_1 + W_8^1 O_1 \quad Y_5 = E_1 - W_8^1 O_1 \quad W_8^1 O_1 = u - jw$

$k = 3 \quad Y_3 = E_3 + W_8^3 O_3 \quad Y_7 = E_3 - W_8^3 O_3 \quad W_8^3 O_2 = -u + jw$

2 real multiplications
2 real additions → shared
Non shared : 4 real addition

$\Rightarrow$ Total 24 real additions, 2 real multiplications

Hence for polyphase FIR w/ FFT, 48 real additions & 34 real multiplications are required

Fixed point arithmetic

$$v_r[m] = \sum_{j=0}^{3} h[8j+r]\, x[8(m-j) + 7 - r] \qquad r = 0 \cdots 7$$

$h[8j+r]\, x[8(m-j)+7-r]$ would need worst case 32-bit signed int

After summation, worst case, $v_r[m]$ would be 34-bit signed int → RNE & saturated to 16-bit for FFT

$$Y_k[m] = \sum_{r=0}^{7} v_r[m]\, e^{-j2\pi kr/8}$$

$Y_k[m] = E_k + W_8^k O_k \quad Y_{k+4}[m] = E_k - W_8^k O_k \quad k = 0, 1, 2, 3$

Stage-1, worst case requires 17-bits

Stage-2, Worst case requires 18-bits

Final stage

k = even, worst case requires 19-bits & $d_0 \pm d_1$ worst case require 18-bits

For min twiddle word length,

let ideal output be Y & hardware produce $\hat{Y}$ before rounding & $|\hat{Y} - Y| \le \varepsilon_Y$

$|\text{round}(\hat{Y}) - \text{round}(Y)| \le \underbrace{\frac{1}{2}}_{\text{hardware rounding}} + |Y - \hat{Y}| + \underbrace{\frac{1}{2}}_{\text{ref rounding}} = \varepsilon_Y + 1$ by triangular inequality

We need $\varepsilon_y + 1 < 3 \Rightarrow \varepsilon_y < 2$ as left side is an integer $\quad y = x/8 \quad \hat{y} = \hat{x}/8 \Rightarrow \varepsilon_x = 8\varepsilon_y$

We cannot store $c = 1/\sqrt{2}$ exactly

Let $\hat{c} = c/2^B$ where $c \in \mathbb{Z}$ & B is the no. of fractional bits

Let $\delta = \hat{c} - c \Rightarrow \hat{c} = c + \delta \quad \delta \rightarrow$ 18-bit signed integer

$s\hat{c} = sc + s\delta \Rightarrow$ Error due to coeff quantization is $s\delta$

But hardware does not calculate $s\hat{c}$, it calculates sc then shifts right by B

$\Rightarrow \hat{p} = \left\lfloor \frac{sc}{2^B} \right\rfloor$ is their truncated part $\Rightarrow \eta = \hat{p} - s\hat{c} \Rightarrow -1 < \eta \leq 0$

$\Rightarrow |\hat{p} - sc| \leq |s||\delta| + |\eta| \quad \Rightarrow |\hat{p} - sc| < |s||\delta| + 1$

$|s| \leq 2^{17} \Rightarrow |\hat{p} - sc| < 2^{17}|\delta| + 1 \Rightarrow \varepsilon_x < 2^{17}|\delta| + 1 \Rightarrow \varepsilon_y < 2^{14}|\delta| + 1/8$

For $c = 1/\sqrt{2}$, take $B = 7$ then $2^7/\sqrt{2} = 64\sqrt{2} \Rightarrow$ nearest int $c = 91 \Rightarrow \delta = (91/128) - (1/\sqrt{2})$

$\Rightarrow \varepsilon_y > 2 \Rightarrow$ 7- or less bits are not sufficient to represent $(1/\sqrt{2})$

as decreasing no. of bits decreases precision & increases error

Take $B = 8 \Rightarrow 2^8/\sqrt{2}$ has nearest int $c = 181 \Rightarrow \varepsilon_y \quad 1.36250296$

$\Rightarrow$ Min no. of fractional bits required to represent twiddle is 8 but the integer representing $1/\sqrt{2}$ itself requires 9 signed int bits

$\Rightarrow$ Intermediates $\frac{(d_0 \pm d_1)}{\sqrt{2}}$ need 26-bits signed int

as $181 \times 2^{17} = 23724032$ which is greater that 2^{24} & less than $2^{25}-1$

After shifting by 8, i.e, dividing by 2^8, max is 92672 $\Rightarrow$ 18-bits signed

Hence, final stage can be represented w/ 19-bit word length in worst case

Architecture

The design accepts one input sample per cycle, so a frame period is 8 cycles and each hardware unit can be used at most 8 times per frame. Pipelining introduces concurrency, hence, the latency of a single frame may exceed 8 cycles without reducing throughput.

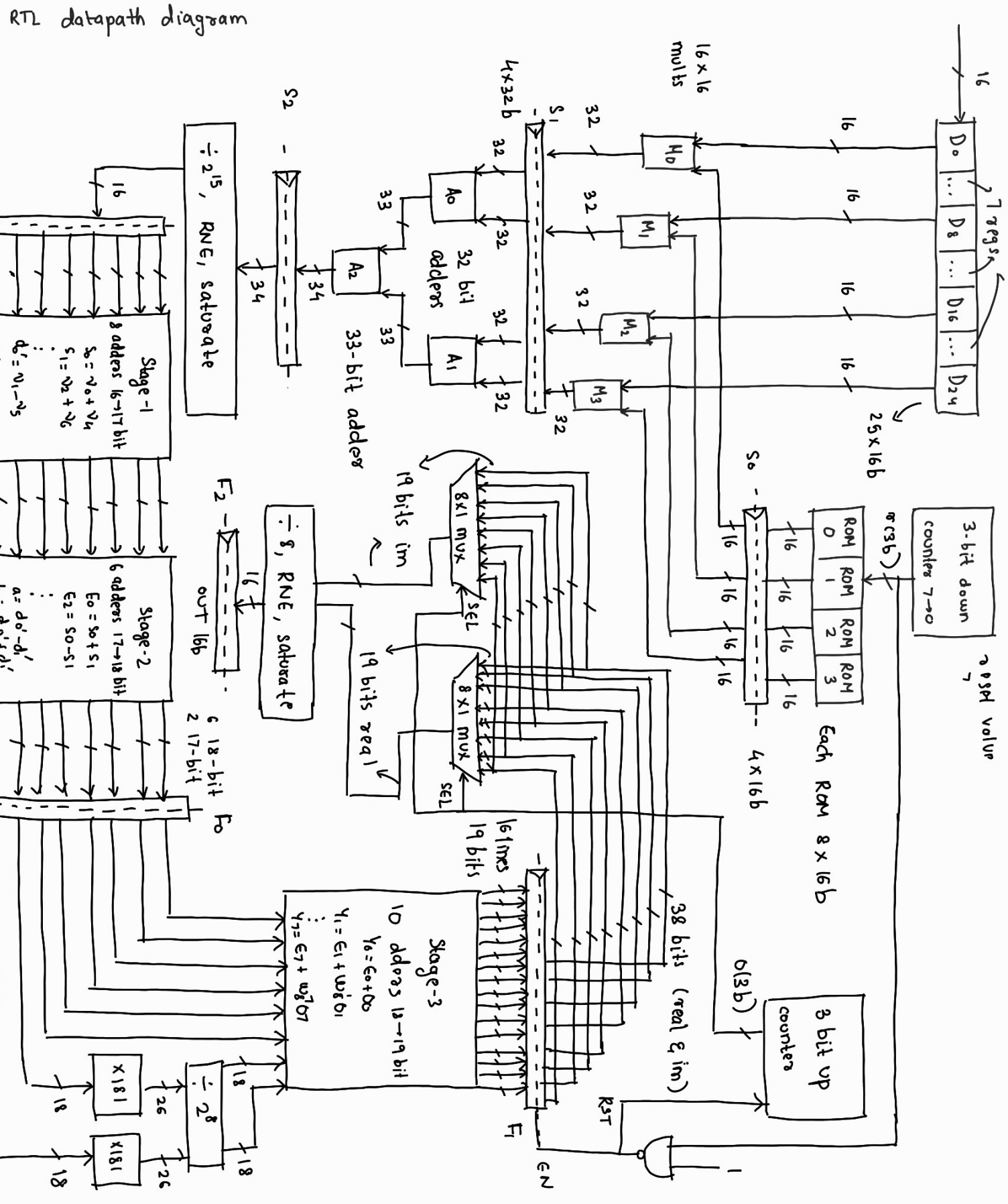
FIR- Total 32 mults and 24 adds, chosen 4 mults, 3 adders

FFT- Total 2 mults 24 adds, chosen 2 mults, 24 adders

Multiplier area grows roughly with n^2 and adder area with n, so multiplier count sets FIR area and the FIR is folded. For slower input (c cycles per sample) the FIR needs $\lceil 4/c \rceil$ multipliers. FFT has only 2 mults and 24 adders (dominate) which are narrow (17 to 19 bit), so folding would add wide multiplexers that cancel the saving, hence, it is left parallel.

Taking the cycle in which the newest sample of a frame is on the input pins as cycle T, the v register is loaded at the end of T+3, F0 at the end of T+4 and F1 at the end of T+5, and the output register at the end of T+6. The first bin Y0 is visible in cycle T+7, so the latency is 7 cycles. Bins Y1 to Y7 follow in cycles T+8 to T+14, and the next frame's Y0 appears at T+15, so the output stream is continuous with one frame every 8 cycles.

Pipeline depth justification- Each stage is limited to one multiplier or a short adder chain. S0: ROM read and delay-line shift. S1: one 16×16 multiplier. S2: two levels of 32/33-bit adders. S3: scaling, round-to-nearest-even and saturation. F0: two levels of 17/18-bit adders. F1: the ×181 constant multiplier (18×9) followed by a single 19-bit adder level. F2: 8:1 mux, ÷8, rounding and saturation. Each stage is expected to fit within the 10 ns period in a 45 nm library. This is an estimate, not a measured result, and it will be confirmed with OpenSTA. Pipelining adds latency but not throughput. If a stage fails timing, a register is added inside it (for example inside the multiplier), at the cost of one cycle of latency each time.



Results dashboard

Note- STA report included w/ the following set of constraints

2 ns : Input delay

10 ns : Clock period

2 ns : Output delay

Worst setup slack : + 6.159 ns

Worst hold slack : + 0.076 ns

Latency : 7 cycles

Worst-case measured error (LSB) over the test suite : 2 LSB